CORE PURE / PROOF BY INDUCTION

Proof by Induction - Matrices

Worked solutions 8 questions · 55 marks

Questions

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1. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}^n = \begin{pmatrix} 2^n & 3^n - 2^n \\ 0 & 3^n \end{pmatrix}$$

[5]

Solution

Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

We prove by induction that, for all $n \in \mathbb{Z}^+$,

$$A^n = \begin{pmatrix} 2^n & 3^n - 2^n \\ 0 & 3^n \end{pmatrix}$$

Base Case: $n = 1$.

$$A^1 = A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

Also,

$$\begin{pmatrix} 2^1 & 3^1 - 2^1 \\ 0 & 3^1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$$

So the result is true when $n = 1$.

Inductive Hypothesis: Assume that for some $k \in \mathbb{Z}^+$,

$$A^k = \begin{pmatrix} 2^k & 3^k - 2^k \\ 0 & 3^k \end{pmatrix}$$

Inductive Step: We must show that this implies

$$A^{k+1} = \begin{pmatrix} 2^{k+1} & 3^{k+1} - 2^{k+1} \\ 0 & 3^{k+1} \end{pmatrix}$$

Starting from $A^{k+1} = A^k A$ and using the inductive hypothesis,

$$\begin{aligned} A^{k+1} &= \begin{pmatrix} 2^k & 3^k - 2^k \\ 0 & 3^k \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 2^k \cdot 2 + (3^k - 2^k) \cdot 0 & 2^k \cdot 1 + (3^k - 2^k) \cdot 3 \\ 0 \cdot 2 + 3^k \cdot 0 & 0 \cdot 1 + 3^k \cdot 3 \end{pmatrix} \\ &= \begin{pmatrix} 2^{k+1} & 2^k + 3(3^k - 2^k) \\ 0 & 3^{k+1} \end{pmatrix} \end{aligned}$$

Now simplify the top right entry:

$$\begin{aligned} 2^k + 3(3^k - 2^k) &= 2^k + 3^{k+1} - 3 \cdot 2^k \\ &= 3^{k+1} - 2 \cdot 2^k \\ &= 3^{k+1} - 2^{k+1} \end{aligned}$$

So

$$A^{k+1} = \begin{pmatrix} 2^{k+1} & 3^{k+1} - 2^{k+1} \\ 0 & 3^{k+1} \end{pmatrix}$$

This is exactly the required form, so the statement is true for $k + 1$.

Therefore, by mathematical induction,

$$\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}^n = \begin{pmatrix} 2^n & 3^n - 2^n \\ 0 & 3^n \end{pmatrix} \quad \text{for all } n \in \mathbb{Z}^+$$

2. Prove by mathematical induction that, for $n \in \mathbb{N}$,

$$\begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}^n = 2^{n-1} \begin{pmatrix} n+2 & -n \\ n & 2-n \end{pmatrix}$$

[6]

Solution

Let

$$A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$$

We will prove by mathematical induction that

$$A^n = 2^{n-1} \begin{pmatrix} n+2 & -n \\ n & 2-n \end{pmatrix} \qquad \text{for all } n \in \mathbb{N}$$

Base Case: $n = 1$

When $n = 1$,

$$A^1 = A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$$

Now evaluate the right-hand side when $n = 1$:

$$2^{1-1} \begin{pmatrix} 1+2 & -1 \\ 1 & 2-1 \end{pmatrix} = 1 \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$$

So the statement is true for $n = 1$.

Induction Hypothesis:

Assume that for some $k \in \mathbb{N}$,

$$A^k = 2^{k-1} \begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix}$$

Inductive Step:

We must show that the result is then true for $n = k + 1$, that is,

$$A^{k+1} = 2^k \begin{pmatrix} (k+1)+2 & -(k+1) \\ k+1 & 2-(k+1) \end{pmatrix}$$

Starting from $A^{k+1} = A^k A$ and using the induction hypothesis,

$$\begin{aligned} A^{k+1} &= A^k A \\ &= \left(2^{k-1} \begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix} \right) \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \\ &= 2^{k-1} \begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \end{aligned}$$

Now multiply the matrices:

$$\begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix}\begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3(k+2)+(-k)(1) & (k+2)(-1)+(-k)(1) \\ 3k+(2-k)(1) & k(-1)+(2-k)(1) \end{pmatrix}$$

$$= \begin{pmatrix} 3k+6-k & -k-2-k \\ 3k+2-k & -k+2-k \end{pmatrix}$$

$$= \begin{pmatrix} 2k+6 & -2k-2 \\ 2k+2 & 2-2k \end{pmatrix}$$

So

$$A^{k+1} = 2^{k-1}\begin{pmatrix} 2k+6 & -2k-2 \\ 2k+2 & 2-2k \end{pmatrix}$$

Factor out 2 from the matrix:

$$A^{k+1} = 2^{k}\begin{pmatrix} k+3 & -(k+1) \\ k+1 & 1-k \end{pmatrix}$$

Since $1-k = 2-(k+1)$, this becomes

$$A^{k+1} = 2^{k}\begin{pmatrix} k+3 & -(k+1) \\ k+1 & 2-(k+1) \end{pmatrix}$$

This is exactly the required form for $n = k+1$.

Therefore, if the statement is true for $n = k$, it is true for $n = k+1$.

Since the statement is true for $n = 1$, and true for $n = k+1$ whenever it is true for $n = k$, it follows by mathematical induction that

$$\begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}^{n} = 2^{n-1}\begin{pmatrix} n+2 & -n \\ n & 2-n \end{pmatrix} \qquad \text{for all } n \in \mathbb{N}$$

This completes the proof.

3. The matrix A is given by

$$A = \begin{pmatrix} 4 & 2 \\ -2 & 0 \end{pmatrix}$$

(a) Prove by induction that

$$A^n = 2^n \begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

for all positive integers n. **[6]**

(b) The matrix B is given by

$$B = (A^n)^{-1}$$

Hence find B in terms of n. **[4]**

Solution

(a) We prove by induction that

$$A^n = 2^n \begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

for all positive integers n.

Base Case: $n = 1$.

The claimed formula gives

$$A^1 = 2^1 \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ -2 & 0 \end{pmatrix} = A$$

So the result is true when $n = 1$.

Inductive Hypothesis: Assume that for some positive integer k,

$$A^k = 2^k \begin{pmatrix} k+1 & k \\ -k & 1-k \end{pmatrix}$$

Inductive Step: We must show that

$$A^{k+1} = 2^{k+1} \begin{pmatrix} k+2 & k+1 \\ -(k+1) & 1-(k+1) \end{pmatrix}$$

Starting from the left-hand side,

$$\begin{aligned} A^{k+1} &= A^k A \\ &= 2^k \begin{pmatrix} k+1 & k \\ -k & 1-k \end{pmatrix} \begin{pmatrix} 4 & 2 \\ -2 & 0 \end{pmatrix} \end{aligned}$$

Now multiply the matrices:

$$\begin{aligned} A^{k+1} &= 2^k \begin{pmatrix} 4(k+1) - 2k & 2(k+1) + 0 \\ -4k - 2(1-k) & -2k + 0 \end{pmatrix} \\ &= 2^k \begin{pmatrix} 2k+4 & 2k+2 \\ -2k-2 & -2k \end{pmatrix} \end{aligned}$$

Factor out 2 from the matrix:

$$A^{k+1} = 2^{k+1}\begin{pmatrix} k+2 & k+1 \\ -(k+1) & -k \end{pmatrix}$$

Since

$$-k = 1 - (k+1)$$

this becomes

$$A^{k+1} = 2^{k+1}\begin{pmatrix} k+2 & k+1 \\ -(k+1) & 1-(k+1) \end{pmatrix}$$

This is exactly the required form, so the statement is true for $k+1$.

Therefore, by induction,

$$A^n = 2^n\begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

for all positive integers n.

(b) Let

$$M = \begin{pmatrix} n+1 & n \\ -n & 1-n \end{pmatrix}$$

so that from part (a),

$$A^n = 2^n M$$

Hence

$$B = (A^n)^{-1} = (2^n M)^{-1} = 2^{-n} M^{-1}$$

Now find M^{-1}. For a 2×2 matrix,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

First calculate the determinant:

$$\begin{aligned} \det(M) &= (n+1)(1-n) - n(-n) \\ &= (n+1)(1-n) + n^2 \\ &= 1 - n^2 + 1 \cdot n - n + n^2 \\ &= 1 \end{aligned}$$

So

$$M^{-1} = \begin{pmatrix} 1-n & -n \\ n & n+1 \end{pmatrix}$$

Therefore

$$B = 2^{-n}\begin{pmatrix} 1-n & -n \\ n & n+1 \end{pmatrix}$$

Hence

$$B = \frac{1}{2^n}\begin{pmatrix} 1-n & -n \\ n & n+1 \end{pmatrix}$$

4. The matrix A is given by

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

Prove by induction that, for $n \geq 1$,

$$A^n = \begin{pmatrix} 2^n & 0 & 0 \\ n\,2^{n-1} & 2^n & 0 \\ \frac{n(n-1)}{2}\,2^{n-2} & n\,2^{n-1} & 2^n \end{pmatrix}$$

[7]

Solution

We prove the result by mathematical induction on n.

Let

$$P(n): \quad A^n = \begin{pmatrix} 2^n & 0 & 0 \\ n\,2^{n-1} & 2^n & 0 \\ \frac{n(n-1)}{2}\,2^{n-2} & n\,2^{n-1} & 2^n \end{pmatrix}$$

Base case: $n = 1$

When $n = 1$, the right-hand side becomes

$$\begin{pmatrix} 2^1 & 0 & 0 \\ 1 \cdot 2^0 & 2^1 & 0 \\ \frac{1(1-1)}{2}\,2^{-1} & 1 \cdot 2^0 & 2^1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} = A$$

So $P(1)$ is true.

Inductive hypothesis

Assume that $P(k)$ is true for some integer $k \geq 1$. That is,

$$A^k = \begin{pmatrix} 2^k & 0 & 0 \\ k\,2^{k-1} & 2^k & 0 \\ \frac{k(k-1)}{2}\,2^{k-2} & k\,2^{k-1} & 2^k \end{pmatrix}$$

Inductive step

We must show that $P(k+1)$ is true.

Starting from

$$A^{k+1} = A^k A$$

Substitute the assumed form for A^k:

$$A^{k+1} = \begin{pmatrix} 2^k & 0 & 0 \\ k\,2^{k-1} & 2^k & 0 \\ \frac{k(k-1)}{2}\,2^{k-2} & k\,2^{k-1} & 2^k \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2^k \cdot 2 & 0 & 0 \\ k\,2^{k-1} \cdot 2 + 2^k \cdot 1 & 2^k \cdot 2 & 0 \\ \frac{k(k-1)}{2}\,2^{k-2} \cdot 2 + k\,2^{k-1} \cdot 1 & k\,2^{k-1} \cdot 2 + 2^k \cdot 1 & 2^k \cdot 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2^{k+1} & 0 & 0 \\ k\,2^k + 2^k & 2^{k+1} & 0 \\ \frac{k(k-1)}{2}\,2^{k-1} + k\,2^{k-1} & k\,2^k + 2^k & 2^{k+1} \end{pmatrix}$$

$$= \begin{pmatrix} 2^{k+1} & 0 & 0 \\ (k+1)2^k & 2^{k+1} & 0 \\ \left(\frac{k(k-1)}{2} + k\right)2^{k-1} & (k+1)2^k & 2^{k+1} \end{pmatrix}$$

Now simplify the bottom-left entry:

$$\left(\frac{k(k-1)}{2} + k\right)2^{k-1} = \left(\frac{k(k-1)+2k}{2}\right)2^{k-1}$$

$$= \left(\frac{k^2 - k + 2k}{2}\right)2^{k-1}$$

$$= \frac{k(k+1)}{2}\,2^{k-1}$$

So

$$A^{k+1} = \begin{pmatrix} 2^{k+1} & 0 & 0 \\ (k+1)2^k & 2^{k+1} & 0 \\ \frac{k(k+1)}{2}\,2^{k-1} & (k+1)2^k & 2^{k+1} \end{pmatrix}$$

This is exactly the required form with $n = k+1$, since

$$(k+1)2^k = (k+1)2^{(k+1)-1}$$

and

$$\frac{k(k+1)}{2}\,2^{k-1} = \frac{(k+1)\big((k+1)-1\big)}{2}\,2^{(k+1)-2}$$

Therefore $P(k+1)$ is true.

Conclusion

Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$, the formula is true for all integers $n \geq 1$:

$$A^n = \begin{pmatrix} 2^n & 0 & 0 \\ n\,2^{n-1} & 2^n & 0 \\ \frac{n(n-1)}{2}\,2^{n-2} & n\,2^{n-1} & 2^n \end{pmatrix}$$

5. The matrix $M = \begin{pmatrix} 3c & 1 \\ 0 & 3c \end{pmatrix}$, where c is a positive constant.

(a) Prove by induction that for all positive integers n,

$$M^n = \begin{pmatrix} (3c)^n & n(3c)^{n-1} \\ 0 & (3c)^n \end{pmatrix}$$

[7]

(b) Given that $\det(M^n) = 144^n$, find the value of c. **[5]**

Solution

(a) We prove the result by induction on n.

Base Case: $n = 1$

$$M^1 = M = \begin{pmatrix} 3c & 1 \\ 0 & 3c \end{pmatrix}$$

Also,

$$\begin{pmatrix} (3c)^1 & 1(3c)^{1-1} \\ 0 & (3c)^1 \end{pmatrix} = \begin{pmatrix} (3c)^1 & 1(3c)^0 \\ 0 & (3c)^1 \end{pmatrix} = \begin{pmatrix} 3c & 1 \\ 0 & 3c \end{pmatrix}$$

So the formula is true when $n = 1$.

Inductive Hypothesis: Assume that for some positive integer k,

$$M^k = \begin{pmatrix} (3c)^k & k(3c)^{k-1} \\ 0 & (3c)^k \end{pmatrix}$$

Inductive Step: We must show that this implies

$$M^{k+1} = \begin{pmatrix} (3c)^{k+1} & (k+1)(3c)^k \\ 0 & (3c)^{k+1} \end{pmatrix}$$

Starting from the inductive hypothesis,

$$\begin{aligned} M^{k+1} &= M^k M \\ &= \begin{pmatrix} (3c)^k & k(3c)^{k-1} \\ 0 & (3c)^k \end{pmatrix} \begin{pmatrix} 3c & 1 \\ 0 & 3c \end{pmatrix} \\ &= \begin{pmatrix} (3c)^k \cdot 3c + k(3c)^{k-1} \cdot 0 & (3c)^k \cdot 1 + k(3c)^{k-1} \cdot 3c \\ 0 \cdot 3c + (3c)^k \cdot 0 & 0 \cdot 1 + (3c)^k \cdot 3c \end{pmatrix} \\ &= \begin{pmatrix} (3c)^{k+1} & (3c)^k + 3ck(3c)^{k-1} \\ 0 & (3c)^{k+1} \end{pmatrix} \end{aligned}$$

Now simplify the top right entry:

$$\begin{aligned} (3c)^k + 3ck(3c)^{k-1} &= (3c)^k + k(3c)^k \\ &= (k+1)(3c)^k \end{aligned}$$

Hence

$$M^{k+1} = \begin{pmatrix} (3c)^{k+1} & (k+1)(3c)^k \\ 0 & (3c)^{k+1} \end{pmatrix}$$

So the statement is true for $k+1$.

Since it is true for $n=1$, and true for $k+1$ whenever it is true for k, it follows by mathematical induction that for all positive integers n,

$$M^n = \begin{pmatrix} (3c)^n & n(3c)^{n-1} \\ 0 & (3c)^n \end{pmatrix}$$

(b) From part (a),

$$M^n = \begin{pmatrix} (3c)^n & n(3c)^{n-1} \\ 0 & (3c)^n \end{pmatrix}$$

This is an upper triangular matrix, so its determinant is the product of its diagonal entries:

$$\begin{aligned} \det(M^n) &= (3c)^n \cdot (3c)^n \\ &= (3c)^{2n} \\ &= (9c^2)^n \end{aligned}$$

We are given that

$$\det(M^n) = 144^n$$

So

$$(9c^2)^n = 144^n$$

Hence

$$9c^2 = 144$$

Therefore

$$c^2 = 16$$

Since c is positive,

$$c = 4$$

6. For $n \in \mathbb{Z}^+$, show, using mathematical induction, that

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}^n = \begin{pmatrix} (-1)^n & -n(-1)^n & \frac{n(n-1)}{2}(-1)^n \\ 0 & (-1)^n & -n(-1)^n \\ 0 & 0 & (-1)^n \end{pmatrix} \quad \textbf{[5]}$$

Solution

Let

$$A = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

and let $P(n)$ be the statement

$$A^n = \begin{pmatrix} (-1)^n & -n(-1)^n & \dfrac{n(n-1)}{2}(-1)^n \\ 0 & (-1)^n & -n(-1)^n \\ 0 & 0 & (-1)^n \end{pmatrix}$$

Base Case: $n = 1$

Substituting $n = 1$ into the right-hand side gives

$$\begin{pmatrix} (-1)^1 & -1(-1)^1 & \dfrac{1(1-1)}{2}(-1)^1 \\ 0 & (-1)^1 & -1(-1)^1 \\ 0 & 0 & (-1)^1 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix} = A$$

So $P(1)$ is true.

Induction Hypothesis: Assume $P(k)$ is true for some $k \in \mathbb{Z}^+$

That is,

$$A^k = \begin{pmatrix} (-1)^k & -k(-1)^k & \dfrac{k(k-1)}{2}(-1)^k \\ 0 & (-1)^k & -k(-1)^k \\ 0 & 0 & (-1)^k \end{pmatrix}$$

We must show that $P(k+1)$ is true.

Inductive Step:

For simplicity, let

$$a = (-1)^k, \qquad b = -k(-1)^k, \qquad c = \frac{k(k-1)}{2}(-1)^k$$

Then

$$A^k = \begin{pmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{pmatrix}$$

Now multiply by A:

$$A^{k+1} = A^k A = \begin{pmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} -a & a-b & b-c \\ 0 & -a & a-b \\ 0 & 0 & -a \end{pmatrix}$$

Now simplify each entry.

First,

$$-a = -(-1)^k = (-1)^{k+1}$$

Next,

$$\begin{aligned} a - b &= (-1)^k - \big(-k(-1)^k\big) \\ &= (-1)^k + k(-1)^k \\ &= (k+1)(-1)^k \\ &= -(k+1)(-1)^{k+1} \end{aligned}$$

Finally,

$$\begin{aligned} b - c &= -k(-1)^k - \frac{k(k-1)}{2}(-1)^k \\ &= \left(-k - \frac{k(k-1)}{2}\right)(-1)^k \\ &= -\frac{2k + k(k-1)}{2}(-1)^k \\ &= -\frac{k^2 + k}{2}(-1)^k \\ &= -\frac{k(k+1)}{2}(-1)^k \\ &= \frac{k(k+1)}{2}(-1)^{k+1} \end{aligned}$$

So

$$A^{k+1} = \begin{pmatrix} (-1)^{k+1} & -(k+1)(-1)^{k+1} & \frac{k(k+1)}{2}(-1)^{k+1} \\ 0 & (-1)^{k+1} & -(k+1)(-1)^{k+1} \\ 0 & 0 & (-1)^{k+1} \end{pmatrix}$$

Since

$$\frac{k(k+1)}{2} = \frac{(k+1)\big((k+1)-1\big)}{2}$$

this is exactly the required form for $n = k+1$.

Therefore $P(k+1)$ is true.

Conclusion:

$P(1)$ is true, and assuming $P(k)$ is true leads to $P(k+1)$ being true. Hence, by mathematical induction,

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}^n = \begin{pmatrix} (-1)^n & -n(-1)^n & \frac{n(n-1)}{2}(-1)^n \\ 0 & (-1)^n & -n(-1)^n \\ 0 & 0 & (-1)^n \end{pmatrix} \quad \text{for all } n \in \mathbb{Z}^+$$

7. The Fibonacci sequence (F_n) is defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$.
You are given that

$$M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

Prove by induction that

$$M^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$$

for all integers $n \geq 1$. **[5]**

Solution

Let $P(n)$ be the statement

$$M^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$$

for integers $n \geq 1$.

Base Case: $n = 1$**.**

We have

$$M^1 = M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

Also,

$$F_2 = F_1 + F_0 = 1 + 0 = 1$$

So

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix}$$

Hence

$$M = \begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix}$$

so $P(1)$ is true.

Inductive Hypothesis.

Assume that for some integer $k \geq 1$,

$$M^k = \begin{pmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{pmatrix}$$

Inductive Step.

We must show that $P(k+1)$ is true.

Starting from $M^{k+1} = M^k M$,

$$\begin{aligned} M^{k+1} &= M^k M \\ &= \begin{pmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} F_{k+1} \cdot 1 + F_k \cdot 1 & F_{k+1} \cdot 1 + F_k \cdot 0 \\ F_k \cdot 1 + F_{k-1} \cdot 1 & F_k \cdot 1 + F_{k-1} \cdot 0 \end{pmatrix} \\ &= \begin{pmatrix} F_{k+1} + F_k & F_{k+1} \\ F_k + F_{k-1} & F_k \end{pmatrix} \end{aligned}$$

Using the Fibonacci recurrence,

$$F_{k+2} = F_{k+1} + F_k \qquad \text{and} \qquad F_{k+1} = F_k + F_{k-1}$$

So

$$M^{k+1} = \begin{pmatrix} F_{k+2} & F_{k+1} \\ F_{k+1} & F_k \end{pmatrix}$$

This is exactly the required form with $n = k + 1$. Therefore $P(k + 1)$ is true.

Conclusion.

Since $P(1)$ is true, and $P(k) \Rightarrow P(k + 1)$, the statement holds for all integers $n \geq 1$ by mathematical induction.

Therefore,

$$M^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \quad \text{for all integers } n \geq 1$$

8.

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Prove by induction that, for all positive integers n,

$$\mathbf{A}^n = \begin{pmatrix} 1 & 2^n - 1 & 1 + (n-2)2^{n-1} \\ 0 & 2^n & n2^{n-1} \\ 0 & 0 & 2^n \end{pmatrix}$$ **[5]**

Solution

We prove the result by induction on n.

Base case: $n = 1$.

The claimed formula gives

$$\mathbf{A}^1 = \begin{pmatrix} 1 & 2^1 - 1 & 1 + (1-2)2^{1-1} \\ 0 & 2^1 & 1 \cdot 2^{1-1} \\ 0 & 0 & 2^1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix} = \mathbf{A}$$

So the result is true when $n = 1$.

Inductive hypothesis: Assume that for some positive integer k,

$$\mathbf{A}^k = \begin{pmatrix} 1 & 2^k - 1 & 1 + (k-2)2^{k-1} \\ 0 & 2^k & k2^{k-1} \\ 0 & 0 & 2^k \end{pmatrix}$$

Inductive step: We must show that the formula is then true for $n = k + 1$.

Using $\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A}$,

$$\mathbf{A}^{k+1} = \begin{pmatrix} 1 & 2^k - 1 & 1 + (k-2)2^{k-1} \\ 0 & 2^k & k2^{k-1} \\ 0 & 0 & 2^k \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Now multiply row by column:

$$\mathbf{A}^{k+1} = \begin{pmatrix} 1 & 1 + 2(2^k - 1) & (2^k - 1) + 2\big(1 + (k-2)2^{k-1}\big) \\ 0 & 2^{k+1} & 2^k + 2k2^{k-1} \\ 0 & 0 & 2^{k+1} \end{pmatrix}$$

Now simplify each entry:

$$1 + 2(2^k - 1) = 1 + 2^{k+1} - 2 = 2^{k+1} - 1$$

$$\begin{aligned} (2^k - 1) + 2\big(1 + (k-2)2^{k-1}\big) &= 2^k - 1 + 2 + (k-2)2^k \\ &= 1 + (k-1)2^k \end{aligned}$$

$$2^k + 2k2^{k-1} = 2^k + k2^k = (k+1)2^k$$

So

$$\mathbf{A}^{k+1} = \begin{pmatrix} 1 & 2^{k+1} - 1 & 1 + (k-1)2^k \\ 0 & 2^{k+1} & (k+1)2^k \\ 0 & 0 & 2^{k+1} \end{pmatrix}$$

This is exactly the given form with $n = k + 1$, since

$$1 + \big((k+1) - 2\big)2^{(k+1)-1} = 1 + (k-1)2^k$$

and

$$(k+1)2^{(k+1)-1} = (k+1)2^k$$

Therefore the statement is true for $n = k + 1$.

Hence, by mathematical induction, for all positive integers n,

$$\mathbf{A}^n = \begin{pmatrix} 1 & 2^n - 1 & 1 + (n-2)2^{n-1} \\ 0 & 2^n & n2^{n-1} \\ 0 & 0 & 2^n \end{pmatrix}$$